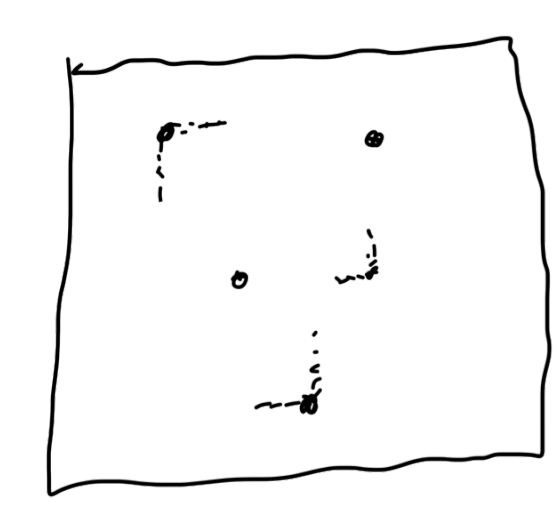


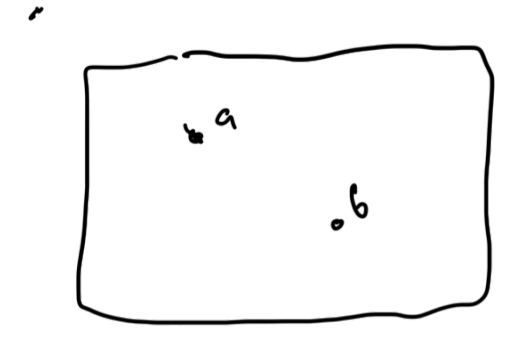
Part 1
Attempt #1

goal: make biggest rect.
possible from 2 pts.



iden: will
"extreme points", i.e.
closest pts. to each corner,
appear in the solution?

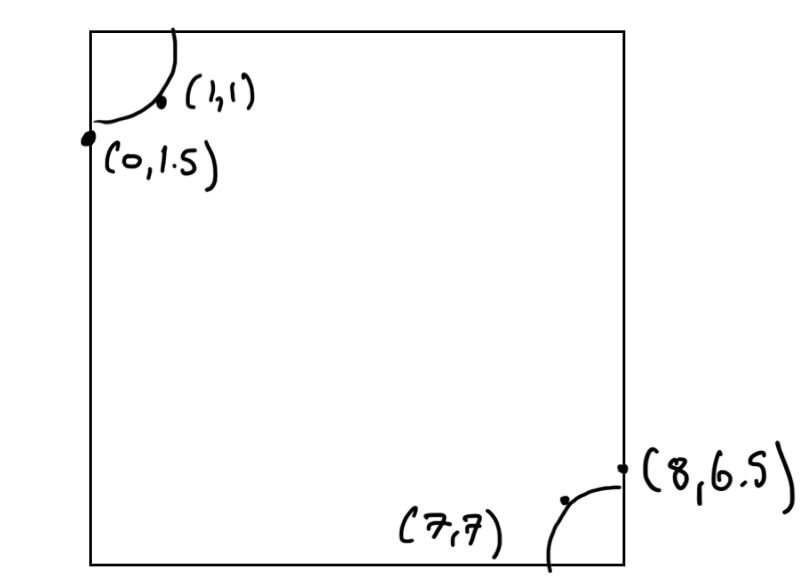
Center:



if there was a c
then was closer to O, O
then a, it would replace a.

examples

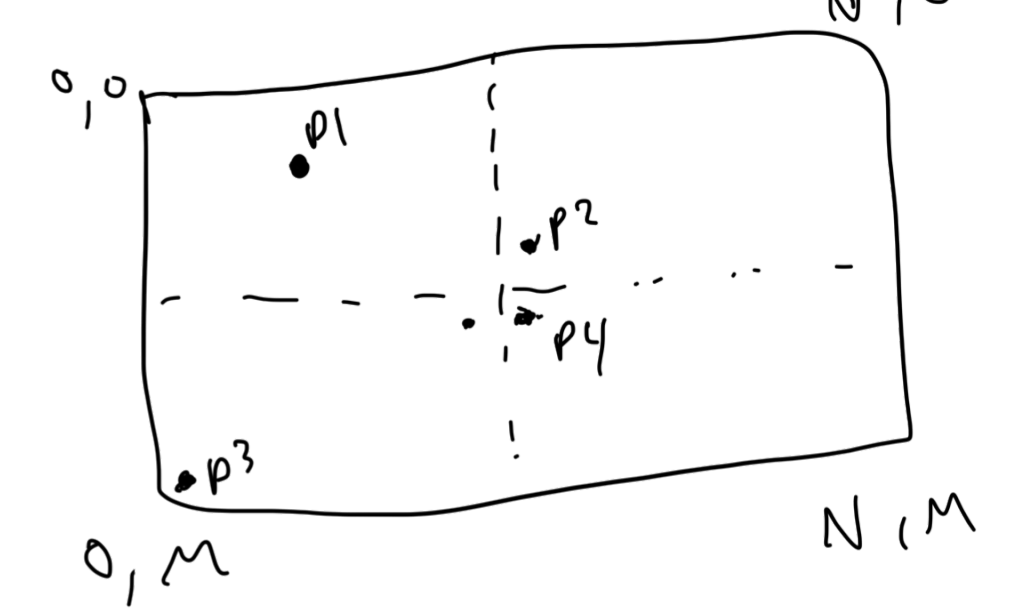
O extreme pt. sol!



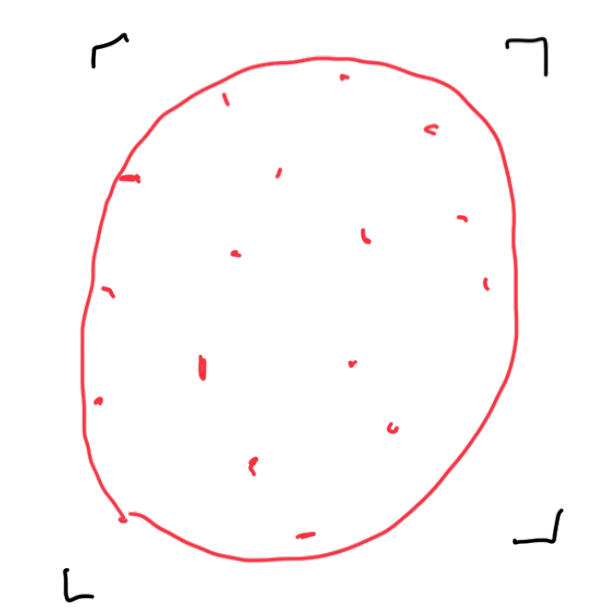
$$8 \times 5 = 40$$

$$6 \times 6 = 36$$

simple 1 extreme pt.-s.



misc



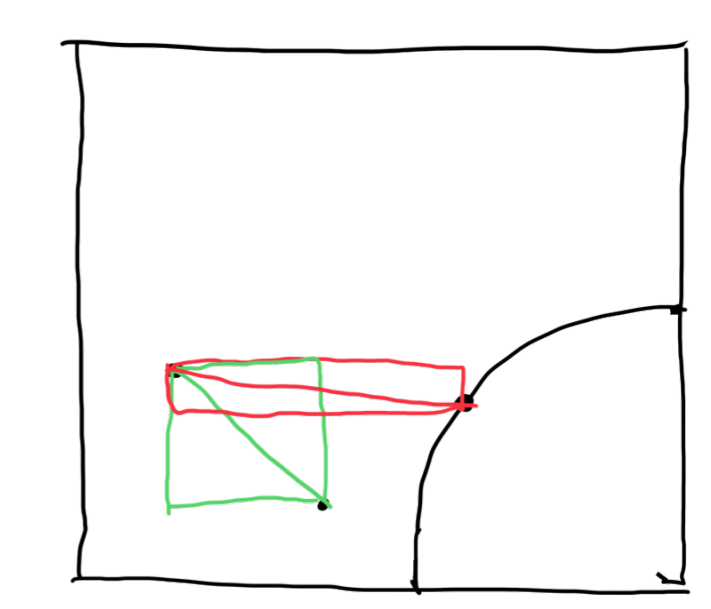
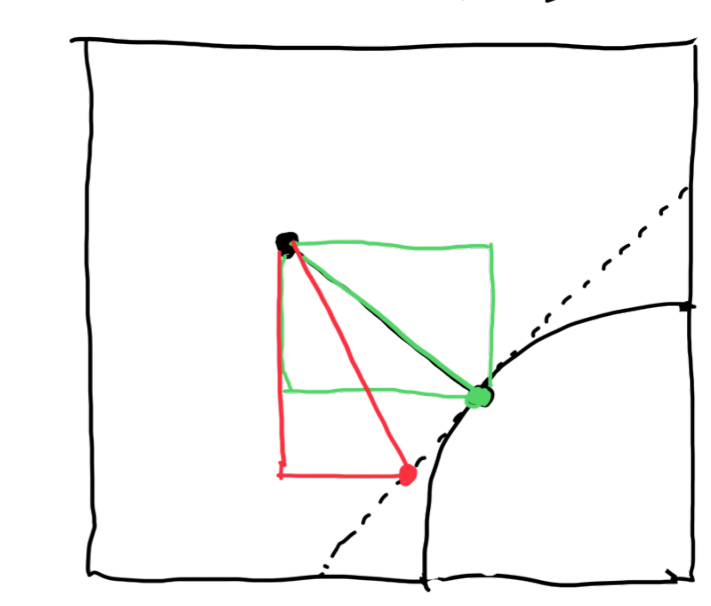
if we start w/ a cloud of
pts, we can draw a circle around
and (probably?) throw out the
points in the middle.

below: angle matters

$$A = \frac{1}{2} \cdot \frac{2c}{\sqrt{2}} \cdot \frac{2c}{\sqrt{2}} = \frac{c^2}{2}$$

$$A = \frac{1}{2} \cdot c \cdot c \cdot \sin 90^\circ = \frac{c^2}{2}$$

- dashed: fixed perimeter (w/ 15' $\Rightarrow$ 14' $\Rightarrow$ 13')
- radius: fixed dist to corner for
any size/shape?



green has way less perimeter
but more area since it's more squar.
(reminder: diag length $\propto$ perimeter)

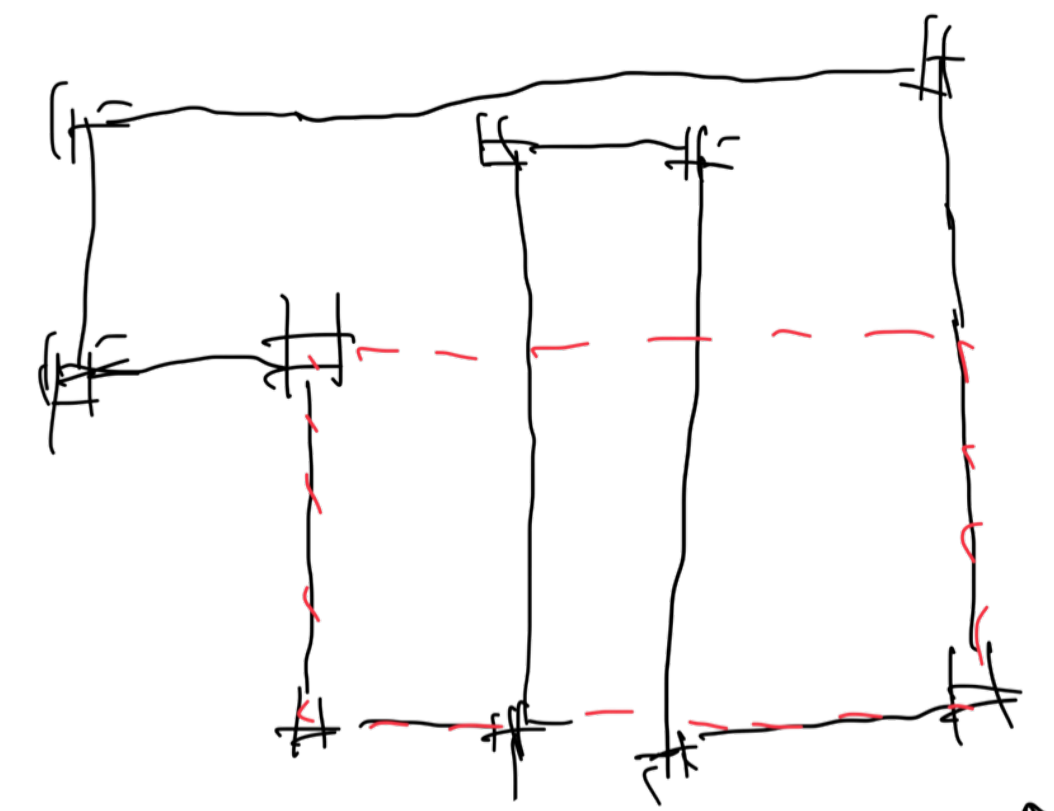
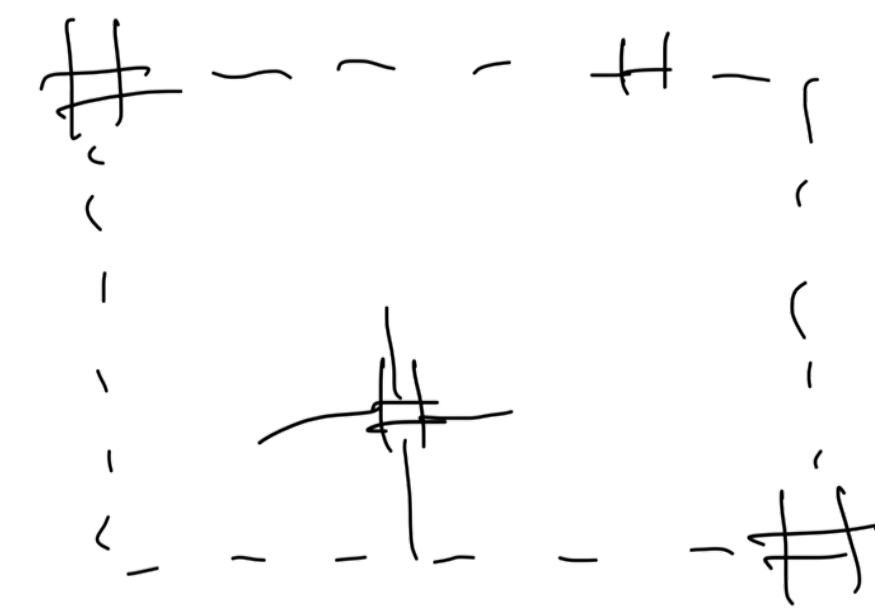
Part 2

check alg: given a pair of pts. that
presume a rect,

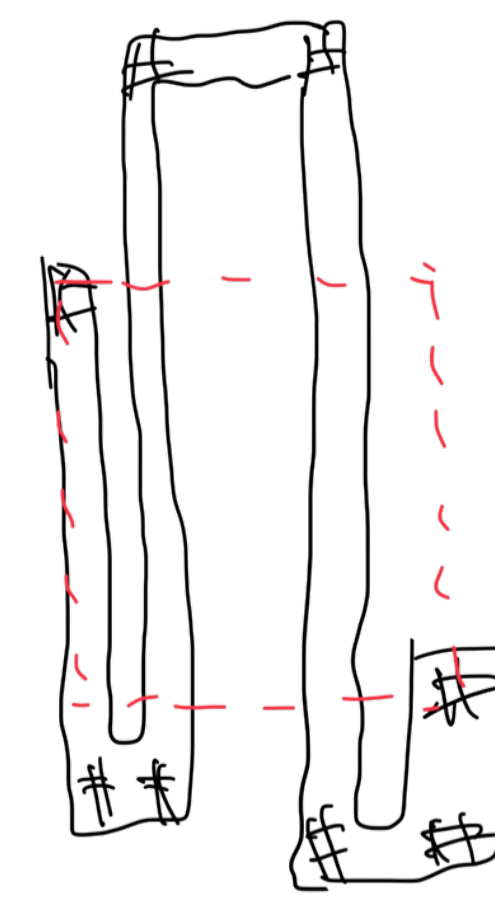
opt 1: start at A and traverse edges. if you ever
"go in" rect, end

opt 2: check all N corners. if any inside rect, end.

correct? ↗



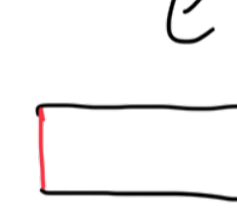
opt 2 breaking case ↗



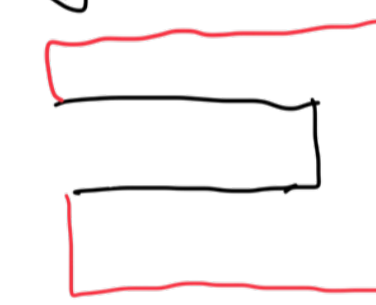
how to know what's in/out?



- You can't know until it's closed.

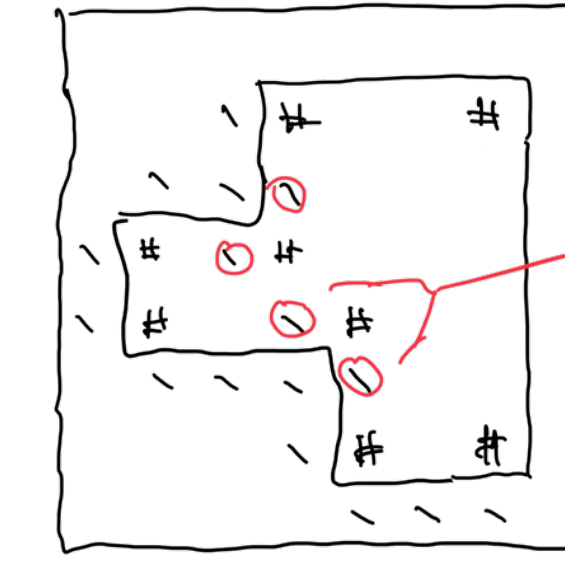


- after it's closed?



- A: from grid edge, find closest lines.

Left Turn Handling



God!

- if next is a L
turn, traverse edge
- it is a L turn, so
turn, traverse edge